Unit 203: Engineering mathematics and science principles

Worksheet 1: Numerical calculations (learner)

Fractions expressed as decimals

Convert the fractions below to decimals:

Examples: $\frac{1}{5}$= 1 ÷ 5 = 0.2, $\frac{11}{20}$ = 11 ÷ 20 = 0.55

1. $\frac{1}{4}$ =
2. $\frac{1}{2}$ =
3. $\frac{31}{4}$ =
4. $\frac{1}{10}$ =
5. $\frac{3}{10}$ =
6. $\frac{7}{10}$ =
7. $\frac{2}{5}$ =
8. $\frac{3}{5}$ =
9. $\frac{4}{5}$ =
10. $\frac{3}{20}$ =
11. $\frac{7}{20}$ =
12. $\frac{19}{20}$ =
13. $\frac{4}{25}$ =
14. $\frac{9}{25}$ =
15. $\frac{23}{25}$ =
16. $\frac{7}{100}$ =
17. $\frac{23}{100}$ =
18. $\frac{106}{200}$ =
19. $\frac{1}{8}$ =
20. $\frac{51}{8}$ =
21. $\frac{9}{40}$ =

Rounding using significant figures

Example: The area of a sheet of aluminium measuring 4.6 cm by 7.2 cm has been calculated as 33.12 cm^2. Since the measurements used in the calculation (4.6 cm and 7.2 cm) are given to two significant figures, the answer should be as well. 33 cm^2 is a more sensible answer.

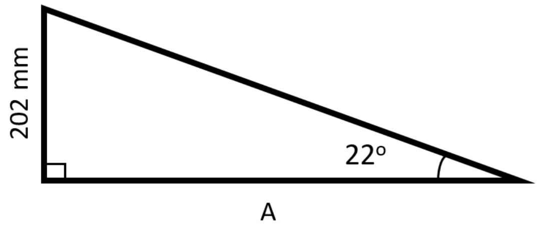
Number	Sig. Fig.	Answer	Number	Sig. Fig.	Answer
456 000	2		0.000567	2	
454 000	2		0.093748	2	
7 981 234	3		0.093748	3	
7 981 234	2		0.093748	4	
1290	2		0.010245	2	
19 602	1		0.02994	2	

Unit 203: Engineering mathematics and science principles

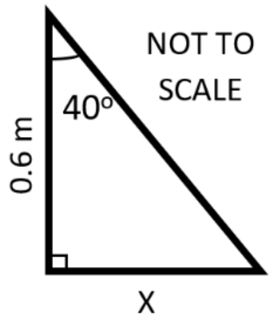
Worksheet 2: Trigonometry (learner)

1. A template is being marked out for a triangular part.
   Calculate the length A to the nearest whole number.

NOT TO SCALE



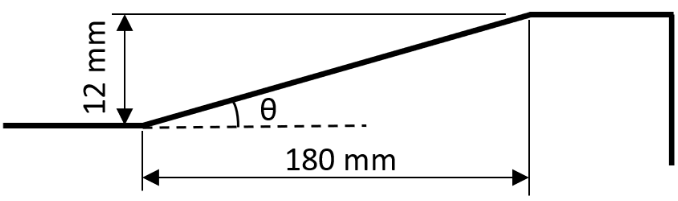
2. The dimensions on the triangle below are needed to program the path of a machine tool.
   Calculate the length of side X to **four** decimal places.



NOT TO SCALE

3. Calculate the taper, θ, on the part shown to **two** decimal places.

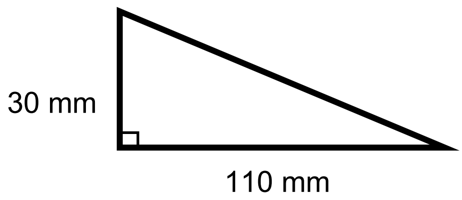
NOT TO SCALE



Unit 203: Engineering mathematics and science principles

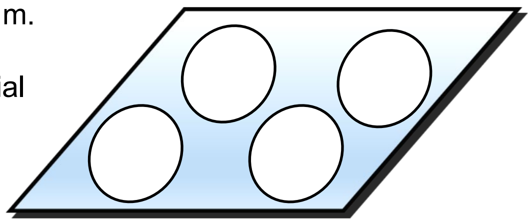
Worksheet 3: Shapes (learner)

1. Calculate area of this triangle.



2. Plastic granules are being stored in a cylinder with a diameter of 500 mm and a height of 900 mm. Calculate the volume of material held in the cylinder, to **three** significant figures.

3. A manufacturer has a sheet of steel that is 1.2 m by 1.4 m. This will be used to cut out four circles of radius 0.3 m. Once the circles have been cut out the remaining material will be scrap.



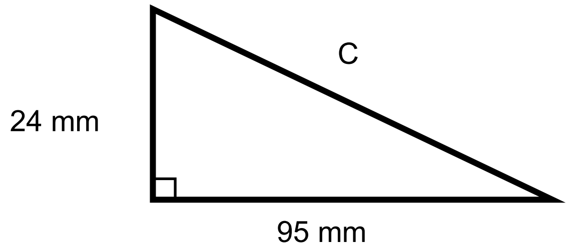
   a) Determine the area of **one** circle.

   b) Calculate the percentage of material that will be scrap.

Unit 203: Engineering mathematics and science principles

Worksheet 4: Pythagoras' theorem (learner)

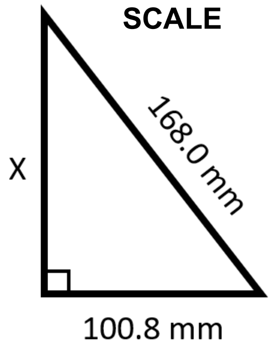
1. A taper is being cut on a lathe. Calculate the length of the taper, C, to the nearest whole number.



NOT TO SCALE

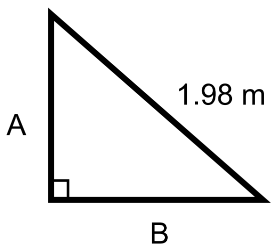
2. A triangular hole needs to be cut in a sheet of metal.
   Calculate the length of side X to **one** decimal place.

NOT TO SCALE



3. A triangular piece of material needs to be marked out for cutting. Calculate the lengths of sides A and B, which are known to be the same length.

NOT TO SCALE



Unit 203: Engineering mathematics and science principles

Worksheet 5: Density (learner)

Remember: $\rho = \frac{m}{V}$

1. A block of material is cuboid in shape, with dimensions 0.3 m × 0.3 m × 0.5 m. The density of the metal is 8000 kg m^{-3}. Calculate the mass of the metal block in kg.

____

____

2. A metal block has a mass of 3510 g. The density of the material is 5.2 g cm^{-3}. Calculate the volume of the block.

____

____

3. A container contains 17600 g of polymer granules. The density of the polymer is 2.2 g cm^3. It is being used to manufacture products that each contain 25 cm^3 of the polymer.
   Calculate the number of products that can be made from the container.

____

____

____

4. A company is making a solid cylinder from recycled plastic. The diameter of the cylinder is 0.2 m, and the density of the plastic is 900 kg m^{-3}.
   The maximum acceptable mass of the cylinder is 27 kg. Calculate the **maximum** acceptable length of the cylinder.

____

____

____

Unit 203: Engineering mathematics and science principles

Worksheet 6: Pressure (learner)

Remember: $P = \frac{F}{a}$

1. The force from an automatic press is 72 kN. The surface area of the press tool that contacts the work piece is 120 mm^2. Calculate the pressure applied by the tool.

2. An output cylinder in a hydraulic system has a radius of 14 mm and experiences a hydraulic pressure of 60 N mm^{-2}.
   Working to **three** significant figures, calculate the force delivered by the cylinder.

3. An industrial press applies a load of 270 kN to press a die into a sheet of material. The area of the die in contact with the material is 4.5×10^{-4} m^2.
   Calculate the pressure that the die applies to the sheet of material.
   Give your answer in N mm^{-2}.

Unit 203: Engineering mathematics and science principles

Worksheet 7: Efficiency (learner)

Remember: Efficiency = (output ÷ input) × 100 %

Complete the tables below to determine the required/missing quantities. These relate to a motor operating at a particular level of efficiency:

Question	**1**	**2**	**3**	**4**
Input	**120 W**	**15 W**	**W**	**1 kW**
Output	**W**	**12 W**	**250 W**	**W**
Efficiency	**80%**	**%**	**50%**	**97%**

Question	**5**	**6**	**7**	**8**
Input	**101 W**	**MW**	**27 W**	**220 W**
Output	**100 W**	**1.5 MW**	**W**	**215 W**
Efficiency	**%**	**75%**	**66.666%**	**%**

Question	9	10	11	12
Input	W	350 W	2 kW	W
Output	450 W	W	1.8 kW	0.9 W
Efficiency	75%	70%	%	40%

Question	13	14	15	16
Input	1500 W	MW	600 W	20 W
Output	1000 W	3 MW	W	15 W
Efficiency	%	75%	45%	%

Use the space below for working out, if needed.

Unit 203: Engineering mathematics and science principles

Worksheet 8: Effect of changes in temperature (learner)

1. Determine the quantity of heat energy required to raise the temperature of 5 kg of steel from 20°C to 800°C. Take the specific heat capacity of steel as 480 J kg^{-1} $°C^{-1}$.

____

____

2. What quantity of heat is necessary to raise the temperature of 12 kg of copper, of specific heat capacity 385 J kg^{-1} $°C^{-1}$, from 20°C to 315°C? Give your answer in MJ.

____

____

3. The specific latent heat required to melt aluminium is 9130 kJ kg^{-1}. Find the quantity of latent heat required to melt 4 kg of aluminium at 660°C.

____

____

4. The specific latent heat required to melt copper is 3850 kJ kg^{-1}. Find the quantity of latent heat required to melt 1.5 kg of copper at 1083°C.

____

____

____

5. An aluminium bar is 200 mm long at 15°C and is heated to 200°C. How long is the bar at 200°C? Take the coefficient of linear expansion of aluminium to be 23×10^{-6} $°C^{-1}$.

____

____

____

____

Unit 203: Engineering mathematics and science principles

Worksheet 9: Material characteristics (learner)

Complete the table below to describe the common engineering metals.

Material	Colour and melting point	Composition	Properties	Uses	Common Forms
Aluminium	Light grey	Pure metal			
Copper					
Brass	980°C	Commonly: 65% copper 35% zinc (but composition varies)			

Material	**Colour and melting point**	**Composition**	**Properties**	**Uses**	**Common Forms**
Low carbon steel					
Medium carbon steel					
High carbon steel					
Cast iron					

Material	**Colour and melting point**	**Composition**	**Properties**	**Uses**	**Common Forms**
Austenitic stainless steel					
Ferritic stainless steel					
Martensitic stainless steel					

Unit 203: Engineering mathematics and science principles

Worksheet 10: Identifying materials (learner)

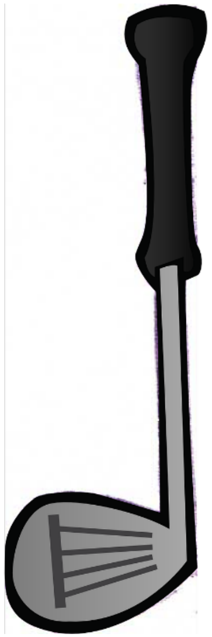
Your tutor will provide you with some samples of material. Carry out investigations to complete the table and identify each material.

Sample	Is it magnetic?	Colour?	Density, g cm^{-3}	Other notes:	I think it is….
1	yes / no				
2	yes / no				
3	yes / no				
4	yes / no				
5	yes / no				

Unit 203: Engineering mathematics and science principles

Worksheet 11: Identifying materials (learner)

Look at the product below. Identify the properties that it requires, giving a reason for each. Suggest a suitable material (or materials).

Product:	**Property needed:**	**Reason:**
 Golf club		

Suggested material: ____

Unit 203: Engineering mathematics and science principles

Worksheet 12: Materials and their properties (learner)

Suggest a material that the product shown below could be made from.
Explain your answer in terms of the properties of the material.

Safety helmet

Material this could be made from: ____

Explanation

____

____

____

____

____

____

____

____

____

____

____